MFGNet: Multibranch Feature Generation Networks for Few-Shot Remote Sensing Scene Classification

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***Abstract*— Few-shot remote sensing scene classification aims to identify unseen classes using only a small number of labeled samples. Considering the large intra-class variances and inter-class similarity of remote sensing scenes, most existing methods focus on feature extraction, ignoring the overfitting problem caused by insufficient samples. To this end, we propose a novel few-shot learning framework, called multibranch feature generation networks (MFGNets), which solves the few-shot scene classification from the source by online sample generation at the representation space. Specifically, we first build a feature generation net to transform the few-shot classification into a regular classification problem, in which the generated samples are achieved by combining the class-specific features with the sampled intra-class features. Then, to ensure the quality of the generated samples, we introduce two novel regularization terms: the intra-class diversity loss (ID-Loss) and the inter-class consistency loss (IC-Loss), which aid the model in generating more diverse samples. Furthermore, we introduce a scale-angle aware self-supervised pretext to learn scale-invariant and rotation-invariant features, improving the model's feature representation capability in remote sensing scenes. We evaluate the proposed method on three publicly available datasets, namely UC_Merced, NWPU-RESISC45, and AID. Our approach has achieved state-of-the-art performance, with an improvement of more than 3.31%, 2.64%, and 6.86% on the most challenging 1-shot tasks, respectively.**

***Index Terms*— Feature generation, few-shot learning, remote sensing scene classification, self-supervised learning.**

I. Introduction

REMOTE sensing scene classification is an essential technology in remote sensing image processing and is widely used in urban planning [1], [2], [3], resource exploration [4], [5], and natural disaster detection [6], [7], [8]. Most mainstream methods currently use convolutional neural networks (CNNs) [9], [10], [11], [12], [13], [14] to classify remote sensing scene, which simply solves the classification problem by combining a feature extractor with a classifier [15]. However, such methods face two dilemmas in practical applications. First, training an effective model requires a large number of data [16], which is expensive and time-consuming. Second, the model lacks excellent generalization capabilities to deal with unseen classes.

To address these issues, few-shot learning [18], [19], [20] has been proposed. Existing few-shot learning methods can be divided into two categories: metric- and optimization-based methods. Metric-based methods [21], [22], [23], [24], [25], [26] start from the idea of clustering that determines sample classes by measuring distance. Optimization-based methods [27], [28], [29] attempt to quickly adjust model parameters by special gradient descent that can be adapted to the new classes under the condition of insufficient data. The direct application of these methods will result in unsatisfactory performance due to the large intra-class variances and inter-class similarity of remote sensing scenes. Therefore, several few-shot learning methods specialized for remote sensing scenes have been proposed, including [30], [31], [32], [33], [34], [35].

Most of these approaches focus on how to extract better features based on metric learning. Cheng et al. [30] proposed a siamese-prototype network that utilizes prototype self-calibration and inter-calibration to acquire more accurate class prototypes. Li et al. [32] proposed an end-to-end network, which adopts the attention technique to automatically discover discriminative regions by delving into the inter-channel and inter-spatial relationships. However, these methods only adjust the network structure and loss function to cope with the challenges posed by large intra-class variances and inter-class similarity under few-shot conditions. The overfitting problem of the classifier still exists due to insufficient samples.

An intuitive solution to address the issue of insufficient samples is to adopt generation-based frameworks [36], [37], which can transform the few-shot classification into a typical classification problem. Some methods [38], [39] that directly model the whole images may lack clarity and diversity of the generated results due to an unstable training process. Unlike these methods, we propose a multibranch feature generation network (MFGNet) that learns the intra-class feature distribution of samples from the feature level. Our method follows the basic assumption [40], [41] that the intra-class feature distributions learned in the seen class, like background, light, noise, and other information can be applied to the unseen class. Specifically, MFGNet learns the class-specific features and intra-class feature distribution separately by decoupling the

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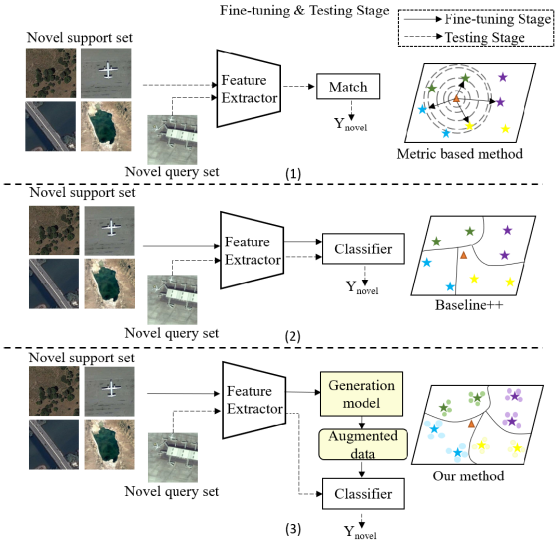

Fig. 1. Fine-tuning (solid lines) and testing processes (dashed lines) of three few-shot classification methods. 1) Metric-based method: the query samples are classified into the nearest support class by calculating the distance. 2) Baseline++ [17]: the classifier is fine-tuned using the support samples, and then the trained classifier is used to predict which class the query sample belongs to. 3) Ours: we augment the samples in the fine-tuning stage with the help of the generation model (indicated by the yellow box), which solves the few-shot classification from the source by generating valuable samples.

sample features and then samples the distribution to obtain the intra-class features. Finally, the newly generated samples are formed by combining the class-specific features with the intra-class features. As shown in Fig. 1, our method stands out from the first two approaches, which rely on either metric learning or fine-tuning the classifier with a limited number of samples. In contrast, we adopt a strategy of generating additional high-quality samples to achieve more accurate and discriminative classification boundaries during both fine-tuning and testing processes.

Considering it is vital to guarantee both the discriminability and diversity of the generated samples [41], [42], our MFGNet first preserves the class information of the original samples, accurately modeling the intra-class feature distribution to ensure the class discriminability. Secondly, to prevent the generated samples from being homogeneous, we design an intraclass diversity loss (ID-Loss) and an interclass consistency loss (IC-Loss). ID-Loss prevents the network from learning only a single intra-class feature distribution while IC-Loss guarantees differences between the generated samples and the original ones. In addition, to further enhance the feature representation of the remote sensing scene, we design a scale-angle aware self-supervised loss (SA-Loss) that captures scale and orientation information, thus boosting few-shot remote sensing scene classification.

The main contributions of our proposed method can be summarized as follows.

1) We introduce an MFGNet to transform few-shot classification into a typical classification problem by feature-level sample generation, significantly improving few-shot classification performance.
2) We design two regularization terms, ID-Loss and IC-Loss, to generate a more discriminative and diverse distribution of samples. ID-Loss helps the model to obtain a richer distribution of intra-class features, while IC-Loss guarantees variability.
3) Considering the characteristics of variable object scale and orientation, a scale-angle aware SA-Loss is designed to enhance the feature representation of remote sensing scenes.

The remaining parts of this article are as follows. Section II reviews related works on few-shot classification and self-supervised learning. Section III mainly shows the detailed method proposed in this article. Section IV displays the experimental results related to the method, and finally, the conclusion is presented in Section V.

II. Related Work

In this section, we mainly discuss some works related to few-shot classification and self-supervised learning.

A. Few-Shot Learning

Existing few-shot learning methods mainly consist of two main categories, metric-, and optimization-based. Most of the metric-based methods achieve few-shot classification by comparing similarities between features. Vinyals et al. [21] proposed matching networks, which introduced attention and memory mechanisms with the requirement that samples within the query set are matched with the most similar samples in the support set. Snell et al. [22] introduced prototypical networks from the idea of clustering, which obtained the class prototypes through the support set. The distance between the query samples and prototypes is calculated to determine which class the samples belong to. Sung et al. [23] proposed a simple end-to-end few-shot classification network and used a nonlinear matching strategy to divide the sample classes of query sets. Unlike the metric learning approach, the optimization-based methods represented by model-agnostic meta-learning (MAML) solved the few-shot problem from the model level. Finn et al. [27] used a meta-learning method that allows the network to quickly adapt to new tasks. Based on [27], Baik et al. [29] designed an adaptive loss named MeTAL, which can be applied to various tasks so as to improve the generalization performance of the network. Chen et al. [17] believed that the use of feature extractors combined with classifiers could also be effective in solving few-shot problems. Some data augmentation-based [38], [39], [41], [42] methods have also been proposed, but currently, none of these methods take into account remote sensing characteristics and it is difficult to maintain a balance of discriminability and diversity when generating samples. In contrast, our article solves the above problem by introducing a self-supervised pretext for remote sensing characteristics and designing a novel multibranch feature generation framework.

B. Remote Sensing Scene Classification

Most of the existing remote sensing scene classification methods are based on deep learning [9], [10], [11], [12],

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[13], [14], the core of them is to obtain a better feature representation. Some methods use pretraining to obtain more discriminative features [9], [10]. Huang et al. [9] proposed a STICL framework to learn spatial-temporally invariant representations in unlabeled data and transfer them to multiple other spatio-temporal scenes. This method allowed the network to produce similar representations of the same view in different spatio-temporal scenarios. There are also approaches [11], [12] that use fusion techniques to improve the capability of the feature representation. To address hard negative classes, Zhang et al. [11] proposed an attribute-cooperated CNN (ACCNN) to exploit attributes as additional guiding information. Some methods introduce attention modules [13], high-frequency features [14], and other ideas in the feature extraction process, which largely improve the performance of remote sensing scene classification. Zhang et al. [14] considered that high-frequency features are important for capturing diversity within classes and similarity between classes and proposed a CNN-based Laplace high-frequency convolutional block to obtain powerful features.

These methods still need a large amount of labeled data for support. To solve the problem of the unseen and limited number of samples in practical applications, few-shot learning is proposed in remote sensing scene classification. Cheng et al. [30] designed a siamese-like network to improve the accuracy of the prototype by adding self-calibration and intercalibration. Cui et al. [31] used a linear classifier designed by PLC to construct a learnable metric module. Li et al. [32] proposed a network named DLA-MatchNet, it added channel attention and spatial attention to Match-Net [21] and designed a variety of methods for fusing attention with the original feature map. Li et al. [33] organized training in a meta way and designed RS-MateNet, using balance loss to maximize the interval between different classes while maintaining the generalization ability of the model. Xiong et al. [35] introduced the semi-supervised learning framework into a few-shot remote sensing classification. It expanded the prediction diversity by using the kernel norm, and designed two additional weighting means to enhance the discriminability of the predictions. Unlike the above methods suffers from model overfitting due to insufficient data, we use a generation model to solve the few-shot scene classification from its source.

C. Self-Supervised Learning

Self-supervised learning [43], [44], [45], [46], which can be trained using unlabeled data, has made great progress in recent years and is also widely used in many areas of remote sensing image processing. Saha et al. [47] proposed a method for multisensor change detection using only the unlabeled target bi-temporal images, training networks in a self-supervised manner through contrast learning. Muhtar et al. [48] used self-supervised learning for semantic segmentation and proposed the IndexNet, which enables the network to learn spatiotemporal invariant features by combining image-level and pixel-level comparisons. In image classification, since labeled samples are difficult to obtain, Yue et al. [49] used a self-supervised learning method with adaptive distillation to train hyperspectral image classification networks. Adaptive soft labels were generated by spatial-spectral similarity measurement. Feng et al. [50] proposed a unique self-supervised adversarial and equivalence network named SAENet, which aims to learn complementary and consistent visual patterns for weakly supervised object detection. SCL-MLNet [34] first applied self-supervised learning to few-shot remote sensing scene classification and enhanced the feature extraction capability of the network by adding unlabeled samples to the training process that built a contrast learning pretext. Different from the above methods, we construct a self-supervised learning task from both target orientation and scale for remote sensing characteristics and design a corresponding loss function to obtain a better feature representation, boosting few-shot classification.

III. Proposed Method

In this section, we describe the MFGNet in detail, including the structure of the MFGNet, the roles of IC-Loss and ID-Loss in sample generation, and the composition of the scale-angle aware self-supervised pretext.

A. Preliminaries Knowledge

To simulate the case of insufficient labeled samples, the general few-shot classification tasks are divided into three processes named training, fine-tuning, and testing. Specifically, given nonoverlapping datasets C_{base} and C_{novel}, the goal is to classify N different classes from C_{novel}, each of which only contains K labeled samples, termed the N-way K-shot setting. The training set D_{train} is constructed from C_{base}, while the test set D_{test} is constructed from C_{novel}. During the training stage, the overall model is trained using all base class samples in D_{train}. In the fine-tuning stage, the feature extractor is frozen, and the classifier is fine-tuned with K labeled samples of the novel class in D_{test}. In the testing stage, the well-trained classifier is used to classify the rest of the unseen samples in D_{test}.

B. Overview

The MFGNet's structure is illustrated in Fig. 2, which consists of a feature extractor and two intra-class feature generation models.

1) In the training stage, the feature extractor plays a crucial role in obtaining the feature map, which is subsequently utilized to extract class-specific features and intra-class features via average pooling and the generation model, respectively. Then, the class-specific features and intra-class features are combined to generate new samples. ID-Loss and IC-Loss are respectively utilized to ensure the diversity between generated and original samples at intra-class and inter-class feature levels. Additionally, to address the variability of remote sensing scenes in terms of object scale and orientation, a scale-angle aware self-supervised pretext is employed to learn rotation-invariant and scale-invariant features.
2) In the fine-tuning stage, the trained feature extractor and intra-class feature generation models are fixed while the

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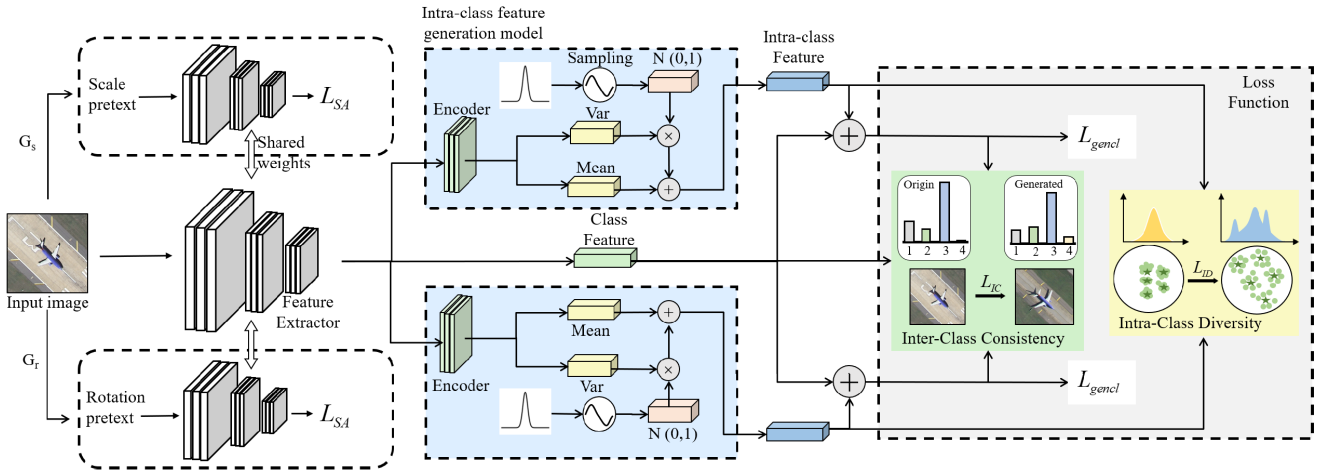


Fig. 2. Framework of the proposed MFGNet. MFGNet consists of a feature extractor and two intra-class feature generation networks. In addition, a self-supervised pretext is designed to capture remote sensing characteristics. Our method learns the intra-class feature distribution of samples with the help of the generation model to achieve data augmentation. IC-Loss and ID-Loss are designed to ensure the variability and distribution diversity of the generated samples. In the testing stage, the few-shot classification problem is transformed into a regular classification problem by multiple sampling of the intra-class feature distribution.

classifier is finetuned with the original and generated samples.

3) In the testing stage, we use the well-trained classifier to classify the unseen samples.

C. Multibranch Feature Generation Network

To alleviate the overfitting problem, MFGNet uses a special sample generation process, which consists of learning class-specific features and sampling the intra-class feature distribution. Since the true intra-class feature distribution is not directly available, we implicitly model it during the sample generation process.

The sample generation is achieved by variational inference, which uses an encoder to fit the posterior distribution $q(z|X)$. The similarity of $q(z|X)$ to the true posterior distribution $p(z|X)$ can be measured by using the Kullback–Leibler divergence, where X denotes the feature maps and z indicates the generated samples

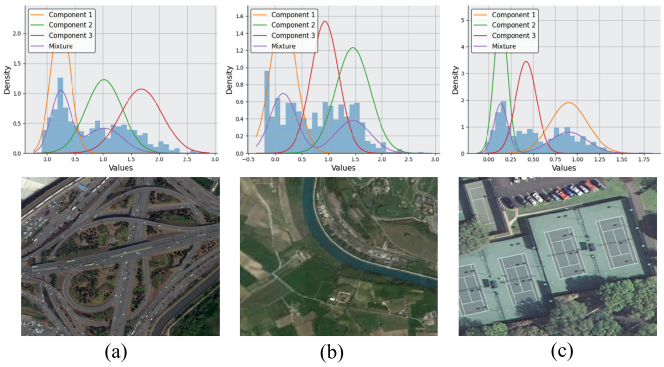
$$D_{\mathrm{KL}}[q(z|X), p(z|X)] = \int_z q(z|X) \log \frac{q(z|X)}{p(z|X)}. \tag{1}$$

Since the sampling process cannot be forward propagated [37], a reparameterization trick is used, which assumes that $p(z|X)$ and $p(z)$ fit the Gaussian distribution. This assumption still holds in the remote sensing scene, because the remote sensing image feature X conforms to the mixed Gaussian distribution (see Fig. 3), which can be decomposed by the Gaussian mixture model (GMM) [51] as a summation of products of Gaussian distributions $p(z)$ and $p(X \mid z)$

$$P(X) = \int_z p(z)p(X \mid z)dz. \tag{2}$$

Obviously, $p(z \mid X)$ also conforms to the Gaussian distribution. Thus, the modeling process can be simplified by learning the mean μ and variance σ of the distribution. A sample ε is drawn from $N(0, 1)$, combining it with the learned mean μ and variance σ results in the intra-class feature z_{in}

$$z_{\mathrm{in}} = \mu + \sigma \odot \varepsilon. \tag{3}$$



(a) (b) (c)

Fig. 3. Feature distribution of three remote sensing scene classification datasets. A Gaussian mixture distribution is adopted to fit them, where the number of Gaussian distribution components is 3. (a) AID. (b) NWPU. (c) UCMerced.

The goal of our model is to obtain a realistic sample distribution $p(X)$, whose objective function can be expressed as $\log p(X)$

$$\log p(X) = l(p, q) + D_{\mathrm{KL}}[q(z|X), p(z|X)] \tag{4}$$

where $l(p, q) = \int_z q(z|X) \log(p(X, z))/(q(z|X))dz$. After the structural transformation, the optimization problem reduces to maximizing $l(p, q)$, which leads to the generation of a more realistic sample distribution $p(X)$ through the use of Bayes' theorem [37]

$$l(p, q) = -D_{\mathrm{KL}}[q(z|X), p(z)] + E_q[\log p(X|z)] \tag{5}$$

where D_{KL} is treated as a regular term and E_q indicates a reconstruction term, the analytical solution for D_{KL} is

$$\begin{aligned} D_{\mathrm{KL}}[q(z|X), p(z)] = {} & D_{\mathrm{KL}}[q(z, X), q(z)p(X)] \\ & + \beta * D_{\mathrm{KL}}\left[q(z), \prod_j q(z_j)\right] \\ & + \sum_j D_{\mathrm{KL}}[q(z_j), p(z_j)]. \end{aligned} \tag{6}$$

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These three terms are called index-code mutual information, total correlation, and dimension-wise KL. As explained in [52] and [53], the total correlation term in (6) plays a crucial role in controlling the posterior probability distribution's independence. By adjusting the hyper-parameter β, this term can effectively enhance the statistical independence, leading to improved model decoupling ability. The overall loss of the intra-class feature distribution modeling process is

$$L_{\text{gen}} = -E_q[\log(p(X|z)] + D_{\text{KL}}[q(z|X), p(z)]. \tag{7}$$

Based on this, the learned intra-class features are summed with the class-specific features to obtain the newly generated samples, and the corresponding features are expressed as z_1 and z_2. This summation operation is beneficial because it models the distribution of intra-class features and maintains the discriminative class-specific features. We employ the classification loss on these newly generated samples. The formula is as follows:

$$L_{\text{gencl}} = \frac{1}{n}\sum_{i=1}^{n} k * \{L_{\text{CE}}(W(z_{1_i}), y_i) + L_{\text{CE}}(W(z_{2_i}), y_i)\} + L_{\text{CE}}(W(z_{0_i}), y_i) \tag{8}$$

where W denotes a classifier, y means the ground truth of the sample, z_1, z_2 presents the generated samples and z_0 means the class-specific feature. k is a hyper-parameter that forces the network to focus more on the generated or original samples by controlling the proportion of classification loss.

D. Diversifying Samples Generation

To obtain more distinguished classification boundaries, we design IC-Loss and ID-Loss to ensure sample diversity from both inter-class and intra-class feature levels, respectively.

1) Inter-Class Consistency Loss: In sample generation, the MFGNet ensures the discriminability of the generated samples by summing the intra-class features with the class-specific features. In using (7) to construct the intra-class feature distribution, E_q indicates a reconstruction item and can be expressed as

$$E_q[\log p(X|z)] = -\frac{1}{n}\sum_{i=1}^{n} ||x_i - z_i||^2 \tag{9}$$

where x_i and z_i denote the original sample feature and generated result, respectively. However, this restrictive constraint can result in insignificant differences, which is contrary to the requirement for variability.

The generated sample exhibits the highest level of variability when it is consistent with the original sample only at the class level. Yun et al. [54] points out that using the consistency of class distribution can effectively address overfitting issues that result in poor generalization and reduce intra-class variance, making it suitable for remote sensing scene classification tasks. To this end, we adopt this idea and design the IC-Loss

$$L_{\text{IC}} = \frac{1}{n}\sum_{i=1}^{n} T^2 * L_{\text{cls}}(x_i, z_i; \theta, T) \tag{10}$$

where $L_{\text{cls}}(x_i, z_i; \theta, T) = D_{\text{KL}}[p(y_i|z_i; \widetilde{\theta}, T), p(y_i|x_i; \theta, T)]$, y means the ground truth of the sample, the temperature parameter T is derived from the idea of knowledge distillation [55], [56]. Its value is related to the class distribution of the generated samples, and the larger the value the flatter the distribution.

With the help of IC Loss, the quality of the reconstructed intra-class features is enhanced. We introduce it into the intra-class distribution learning process, and (7) can be reexpressed as

$$L_{\text{gen}} = L_{\text{IC}} + D_{\text{KL}}[q(z|X), p(z)]. \tag{11}$$

2) Intra-Class Diversity Loss: As the process of sampling involves inherent uncertainty, relying on a single intra-class feature generation model is susceptible to learning incomplete distributions. Even though a multibranch model has been developed, it is still vulnerable to encountering issues related to distribution diversity due to the lack of interaction among branches. To address this limitation and ensure that the learned distributions across different intra-class feature generation models are distinct, we have introduced the ID-Loss into our approach

$$L_{\text{ID}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{1 - \cos(z_{in1_i}, z_{in2_i})}{1 - \cos(m_{1_i}, m_{2_i})}\right] \tag{12}$$

where m_{1_i}, m_{2_i} denote the samples of $N(0, 1)$ distribution obtained by sampling, and z_{in1_i}, z_{in2_i} denote the intra-class features. This term amplifies the distance between intra-class features as much as possible when different branches face similar m_1 and m_2, forcing the generated samples to exhibit greater diversity.

E. Scale-Angle Aware Self-Supervised Pretext

In remote sensing scenes, same class objects often vary in terms of scale and orientation [57], [58]. To deal with this challenge, we design a scale-angle aware self-supervised pretext to improve the robustness of the network by capturing remote sensing scene characteristics. For a set of input images I, they are rotated by $\{i * 90|i \in G_r\}$ degrees by G_r and enlarged by $\{i|i \in G_s\}$ times by G_s, where $G_r = \{0, 1, 2, 3\}$ and $G_s = \{1, 2, 3, 4\}$. Thus the original input images can be transformed into two sets $I_r = \{(I_{r_i}, G_{r_i})|i \in \{0, 1, 2, 3\}\}$ and $I_s = \{(I_{s_i}, G_{s_i})|\ i \in \{0, 1, 2, 3\}\}$. These two groups of constructed I_r and I_s images are then fed into the feature extractor module to generate the features F_s and F_r, respectively. Finally, the rotation angle or scale enlargement is predicted by the perceptual network W_s and W_r, as shown in Fig. 4.

During the training, we develop a scale-angle loss to assist this process

$$L_{\text{SA}} = \frac{1}{n}\sum_{i=1}^{n}\{L_{\text{CE}}(W_r(F_{r_i}), G_{r_i}) + L_{\text{CE}}(W_s(F_{s_i}), G_{s_i})\}. \tag{13}$$

By empowering the network with the capability to adapt different scales and orientations through a scale-angle aware self-supervised pretext, more robust features can be obtained.

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TABLE I
SPLIT ON THREE DATASETS

Datasets	Training Class	Validation Class	Test Class
UC_Merced	Agricultural, Baseball Diamond, Buildings, Chaparral, Dense Residential, Freeway, Harbor, Medium Density Residential, Overpass, Parking lot	Airplane, Forest, Intersection, Runway, Storage Tanks	Beach, Golf Course, Mobile Home Park, River, Sparse Residential, Tennis Courts
NWPU	Airplane, Baseball Diamond, Beach, Bridge, Chaparral, Church, Cloud, Desert, Freeway, Golf Course, Harbor, Island, Lake, Meadow, Mobile Home Park, Mountain, Palace, Railway, Rectangular Farmland, Roundabout, Sea Ice, Ship, Sparse Residential, Stadium, Wetland	Thermal Power Station, Overpass, Railway Station, Tennis Court, Commerical Area, Snowberg, Storge Tank, Runway, Terrace, Industrial Area	Circular Farmland, Basketball Court, Forest, Dence Residential, River, Ground Track Field, Intersection, Medium Residential, Parking Lot, Airport
AID	Airport, Bare land, Baseball field, Beach, Bridge, Commercial, Dense Residential, Desert, Farmland, Meadow, Pond, Port, Railway station, Resort, Stadium	Medium Residential, Mou ntain, Park, Parking, Playground	River, School, Sparse Residen tial, Square, Forest, Industrial, Center, Church, Storage Tanks

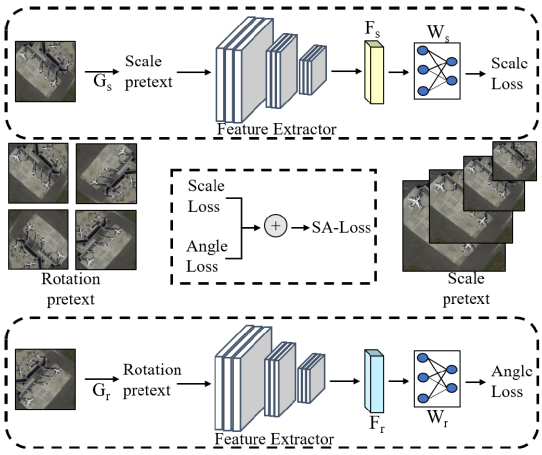

Fig. 4. Process of scale-angle aware self-supervised pretext, scale loss and angle loss are integrated to get SA-loss.

F. Loss Function

The total loss in the MFGNet training stage is defined as

$$L_{\text{total}} = D_{\text{KL}} + L_{\text{gencl}} + \lambda L_{\text{IC}} + \varphi L_{\text{ID}} + \theta L_{\text{SA}} \tag{14}$$

where D_{KL}, L_{IC} [see (11)] and L_{ID} [see (12)] are used in the sample generation process, L_{gencl} [see (8)] represents the classification loss, and L_{SA} [see (13)] plays a role in the self-supervised learning process. λ, φ, θ are three hyper-parameters.

IV. EXPERIMENTS AND ANALYSIS

In this section, we will demonstrate the superior performance of our method in solving few-shot remote sensing scene classification through experiments on three widely used datasets.

A. Datasets

We evaluate model performance using three widely used datasets, named the University of California Merced (UC_Merced) dataset [59], the North-Western Polytechnical University-RESISC45 (NWPU) dataset [10], and the Aerial Image Data (AID) dataset [60]. The data splitting method is shown in Table I.

The UC_Merced dataset contains 21 categories, each includes 100 images of size 256 × 256 pixels, which are divided into a training set, a validation set, and a test set, where the training set consists of Agricultural, Baseball Diamond, Buildings, Chaparral, Dense Residential, Freeway, Harbor, Medium Density Residential, Overpass, and Parking Lot. The validation set has Airplane, Forest, Intersection, Runway, and Storage Tanks. The test set including Beach, Golf Course (Golfc), Mobile Home Park (Moblieh), River, Sparse Residential (Sparser), and Tennis Courts (Tennisc).

The NWPU-RESISC45 dataset, published by Northwestern Polytechnical University, is a benchmark for remote sensing scene classification, which contains 45 categories, each of which consists of 700 RGB images of size 256 × 256 pixels, we also divide it into a training set, a validation set, and a test set, where the training set consists of Airplane, Baseball Diamond, Beach, Bridge, Chaparral, Church, Cloud, Desert, Freeway, Golf Course, Harbor, Island, Lake, Meadow, Mobile Home Park, Mountain, Palace, Railway, Rectangular Farmland, Roundabout, Sea Ice, Ship, Sparse Residential, Stadium, and Wetland. The validation set has Thermal Power Station, Overpass, Railway Station, Tennis Court, Commerical Area, Snowberg, Storge Tank, Runway, Terrace, and Industrial Area. The test set including Circular Farmland (Circularf), Basketball Court (Baseket), Forest, Dence Residential (Denser), River, Ground Track Field (Groundt), Intersection (Inters), Medium Residential (Mediumr), Parking Lot (Parkingl), and Airport.

The AID dataset, jointly published by Huazhong University of Science and Technology and Wuhan University, contains 30 categories with varying numbers of samples in each category, ranging from 220 to 420, all of which are RGB images of 600 × 600 pixels in size. We divide it into a training set, a validation set, and a test set, where the training set consists of Airport, Bare land, Baseball field, Beach, Bridge, Commercial, Dense Residential, Desert, Farmland, Meadow, Pond, Port, Railway station, Resort, and Stadium. The validation set has Medium Residential, Mountain, Park, Parking, and Playground. The test set includes River, School, Sparse Residential (Sparser), Square, Forest, Industrial (Indus), Center, Church, Storage Tanks (Storaget), and Viaduct.

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All of the above datasets are split in a manner consistent with [31].

B. Experiment Settings

In the training stage, all base class samples are used to train the overall model. During the fine-tuning stage, to simulate the situation of insufficient data, we randomly select N classes from the novel classes and then select K samples from each class to form the support set. This task composition is called N-way K-shot. To be consistent with the comparison experiments [17], [21], [22], [23], [27], [31], [32], [33], [34], both 5-way 1-shot and 5-way 5-shot settings are used in the specific experimental process. In the testing stage, for each N-way K-shot task, we randomly select 15 remaining samples in each class to form the query set. We train the classifier using the support set samples, and repeat 200 iterations, after which the trained classifier is used to classify the query set samples. The overall evaluation process was repeated 600 times to reduce the instability associated with sample selection.

For a fair comparison, the feature extraction network used in our article is Resnet12 [61], which is the same as [31]. In the training stage, we use the Adam optimizer [62], the learning rate of the feature extraction and classifier is set to 0.001, and for the rest of the network structure, it is set to 0.0001. The hyper-parameters λ, φ, θ are each set to 1. We also use Mixup [63] in the feature extraction process. In the fine-tuning stage, the number of samples for the $N(0, 1)$ distribution is set to 4, which means that each intra-class feature generation model generates four new samples. The parameter k is set to 1.5. Due to the limited number of samples in the UC_Merced dataset, the parameter a is set to 5.0 for better training and is set to 5.5 in the remaining datasets.

C. Comparison With Other Few-Shot Classification Methods

The existing few-shot classification methods are mainly divided into metric- and optimization-based methods. Based on this, some methods specifically for remote sensing scenes have also achieved good performance. We demonstrate the effectiveness of our approach by comparing it with a series of the state-of-the-art methods, such as Matching Networks [21], Prototypical Networks [22], Relation Networks [23], MAML [27], Baseline++ [17], remote sensing MetaNet (RS-MetaNet) [33], discriminative learning of adaptive match network (DLA-MatchNet) [32], self-supervised contrastive learning-based metric learning network (SCL-MLNet) [34], and metakernel networks (MKN) [31].

As shown in Table II, we can find that our method achieves the state-of-the-art performance, in terms of both 1-shot and 5-shot settings in the NWPU dataset. The following conclusions can be obtained. First, our MFGNet has raised the baseline (Baseline++ [17]) by 11.08%/9.65% in case of 1-shot/5-shot. Second, our proposed method have surpassed all the metric-based methods and optimization-based method, it beats the best metric-based method MKN by about 6.86%/3.77% for 1-shot/5-shot setting. Third, our method obtains a more significant boost at the 1-shot setting, this great improvement convincingly substantiates the effectiveness of the generation model when the labeled test sample is severely inadequate.

TABLE II

FEW-SHOT CLASSIFICATION ACCURACY (%) COMPARISON ON THE NWPU DATASET. THE BOLD VALUES REPRESENT THE BEST ACCURACY AMONG THESE METHODS IN EACH CASE

Method	5-way	
	1-shot	**5-shot**
Matching Networks [21]	37.28 ± 0.99	52.31 ± 0.73
Prototypical Networks [22]	43.99 ± 0.78	69.35 ± 0.66
Relation Networks [23]	44.55 ± 0.79	68.21 ± 1.39
MAML [27]	42.29 ± 0.76	61.85 ± 0.81
Baseline++ [17]	61.62 ± 0.74	76.79 ± 0.94
RS-MetaNet [33]	52.78 ± 0.09	71.49 ± 0.81
DLA-MatchNet [32]	60.80 ± 0.70	77.63 ± 0.46
SCL-MLNet [34]	62.21 ± 1.12	80.86 ± 0.76
MKN [31]	65.84 ± 0.89	82.67 ± 0.55
Ours	**72.70 ± 1.15**	**86.44 ± 0.36**

TABLE III

FEW-SHOT CLASSIFICATION ACCURACY (%) COMPARISON ON THE AID DATASET. THE BOLD VALUES REPRESENT THE BEST ACCURACY AMONG THESE METHODS IN EACH CASE

Method	5-way	
	1-shot	**5-shot**
Matching Networks [21]	41.53 ± 0.54	50.04 ± 0.34
Prototypical Networks [22]	49.80 ± 0.56	68.64 ± 0.29
Relation Networks [23]	53.51 ± 0.68	67.39 ± 1.04
MAML [27]	47.93 ± 0.72	61.79 ± 0.75
Baseline++ [17]	44.68 ± 0.63	61.35 ± 0.88
RS-MetaNet [33]	56.32 ± 0.55	74.48 ± 1.11
DLA-MatchNet [32]	53.41 ± 0.99	70.61 ± 0.36
SCL-MLNet [34]	59.46 ± 0.96	76.31 ± 0.68
MKN [31]	57.29 ± 0.59	75.42 ± 0.10
Ours	**59.93 ± 1.08**	**76.42 ± 0.76**

Table III summarizes the results on AID datasets, which lead to similar observations as those in Table II. Our MFGNet also achieves superior performance with an improvement of 2.64%–18.4% in the case of 1-shot and 1%–26.38% in case of 5-shot. It beats the baseline by 15.25%/15.07% in 1-shot/5-shot setting. In addition, the improvement on 1-shot task is more significant than on 5-shot task. This further verifies the excellent performance of our method in solving few-shot problems, especially in the 1-shot setting.

We also performed comparison experiments on the UC_Merced dataset. In Table IV, our MFGNet outperforms baseline 19.34% in 1-shot task and 16.49% in 5-shot task in terms of Acc. It beats the best metric-based methods MKN by about 3.31% in 1-shot setting, while it is slightly inferior to MKN in the 5-shot task. We speculate the reason is that this dataset is comparatively small. Due to the small scale, it is difficult to fit the intra-feature distribution.

In summary, among all these three datasets, our method reaches significant improvement over baseline and achieves state-of-the-art in most cases. Specifically, when the labeled samples are incredibly sparse in the fine-tuning stage, MFGNet has outstanding properties compared to other methods. The above experiments confirm the effectiveness and superiority of the proposed method.

D. How Our Model Solves the Few-Shot Problem

To further understand how our model works, we use the same experiment settings on the Baseline++ [17] (the baseline

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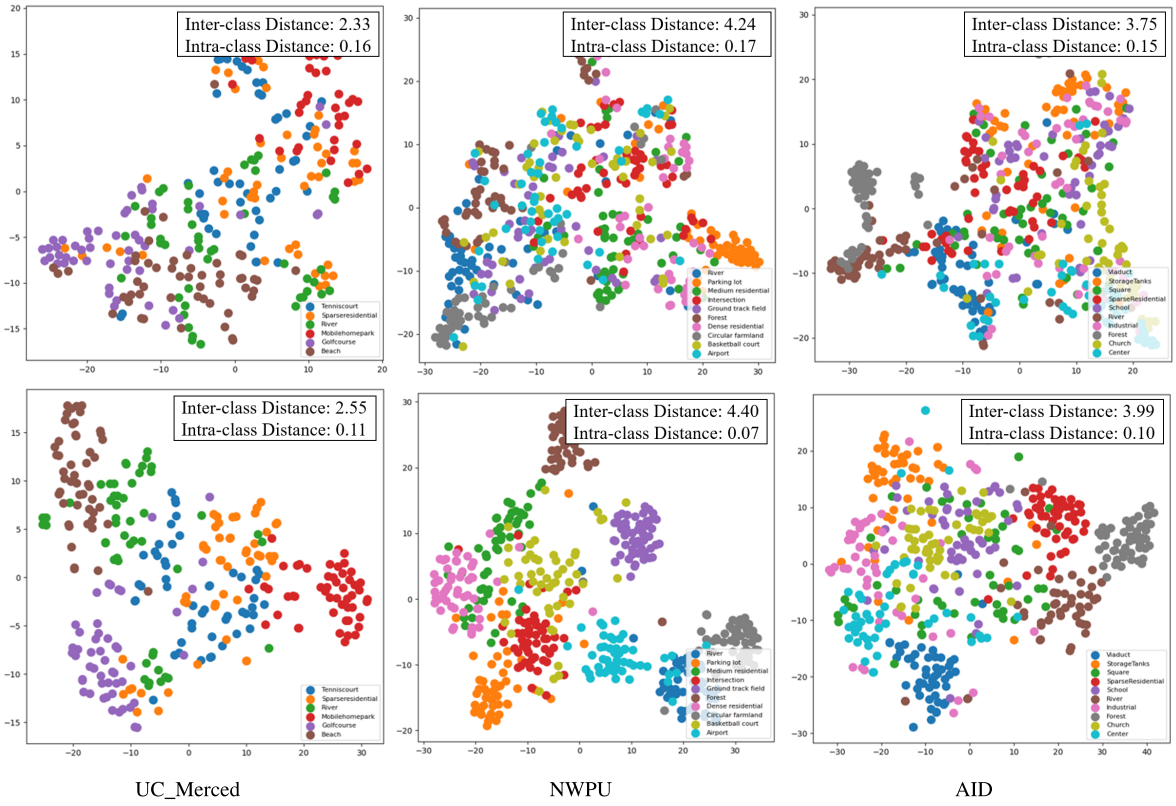

Fig. 5. Comparison of feature extraction ability between our method and baseline model [17] in the face of novel classes. From top to bottom are baseline and our method. The top right corner of each image shows the intra-class and inter-class distance (using European distance) of the samples.

TABLE IV

FEW-SHOT CLASSIFICATION ACCURACY (%) COMPARISON ON THE UC_MERCED DATASET. THE BOLD VALUES REPRESENT THE BEST ACCURACY AMONG THESE METHODS IN EACH CASE

Method	5-way	
	1-shot	**5-shot**
Matching Networks [21]	42.22 ± 1.02	52.32 ± 0.92
Prototypical Networks [22]	50.23 ± 0.61	70.54 ± 0.20
Relation Networks [23]	52.43 ± 0.87	68.65 ± 0.95
MAML [27]	47.53 ± 0.71	63.13 ± 0.92
Baseline++ [17]	42.42 ± 0.46	60.06 ± 0.54
RS-MetaNet [33]	57.23 ± 0.56	76.08 ± 0.28
DLA-MatchNet [32]	53.76 ± 0.62	63.01 ± 0.51
SCL-MLNet [34]	51.37 ± 0.79	68.09 ± 0.92
MKN [31]	58.45 ± 0.54	**77.92 ± 0.32**
Ours	**61.76 ± 0.59**	76.55 ± 0.40

of our article) to explore the model feature extraction capability and classification performance. Tables II–IV show the performance comparison between our model and baseline in the 1-shot and 5-shot cases. Due to the introduction of the generation model and self-supervised learning, our method brings substantial improvement. Especially in the UC_Merced dataset, the use of self-supervised learning and generated samples alleviates the performance deficiency of the feature extraction module. In addition, the advantage of our method in the 1-shot task can also be found.

We visualized the features extracted by these two methods. Fig. 5 shows the distribution of the features when facing novel samples in the UC_Merced, NWPU, and AID datasets. It can be found that the baseline does not distinguish the class features well when facing novel classes. Especially, in NWPU and AID datasets, there are even cases of mixing different class samples, which can lead to serious degradation of the classifier. Moreover, the intra-class distance of features has a decrease of 0.05–0.1 compared to the baseline, while the inter-class distance has an increase of 0.16–0.24, which indicates that our method can effectively narrow the intra-class feature distance while increasing the inter-class distances. Our article constructs an MFGNet and designs a self-supervised learning pretext directly related to the characteristics of remote sensing scenes during the training process. Thus, the features obtained are more robust and the classification boundaries between samples are clearer.

Based on this, we plot the confusion matrix of the classification results to further demonstrate that the classifier combined with the generated samples can better handle the few-shot classification. The results are shown in Figs. 6–8. We experimented with 1-shot and 5-shot tasks, and the classification performance of both methods improved as the number of labeled samples increased. In comparison, the diagonal squares in the confusion matrix of MFGNet are darker due to the generated samples in the fine-tuning stage, indicating that the samples are more likely to be correctly classified and the classification boundary is more obvious.

Since our method models the background information and generates new samples via intra-class feature sampling, we do not change the class information of the samples. As depicted

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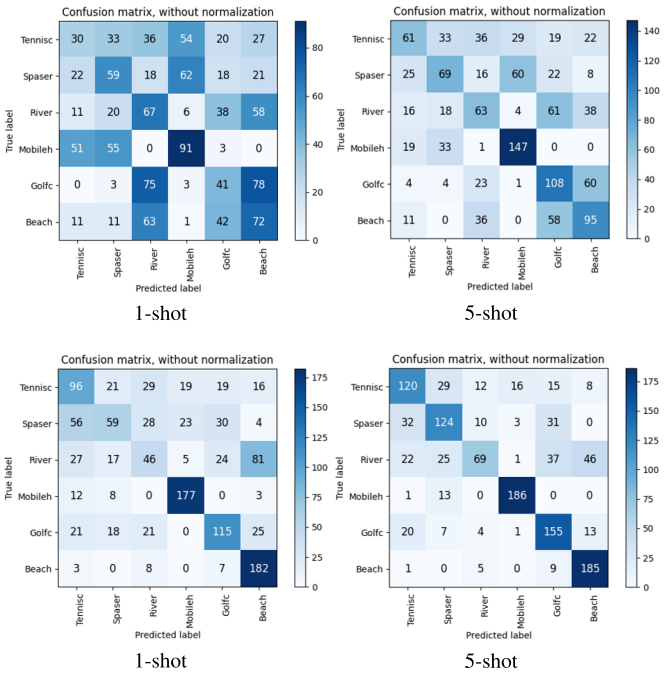

Fig. 6. Confusion matrix for novel classes under the UC_Merced dataset. From top to bottom are baseline [17] and our method.

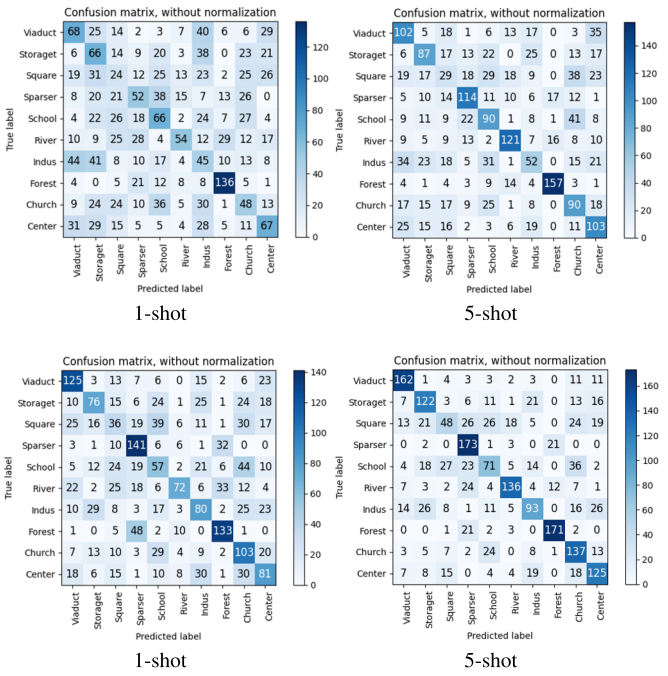

Fig. 8. Confusion matrix for novel classes under the AID dataset. From top to bottom are baseline [17] and our method.

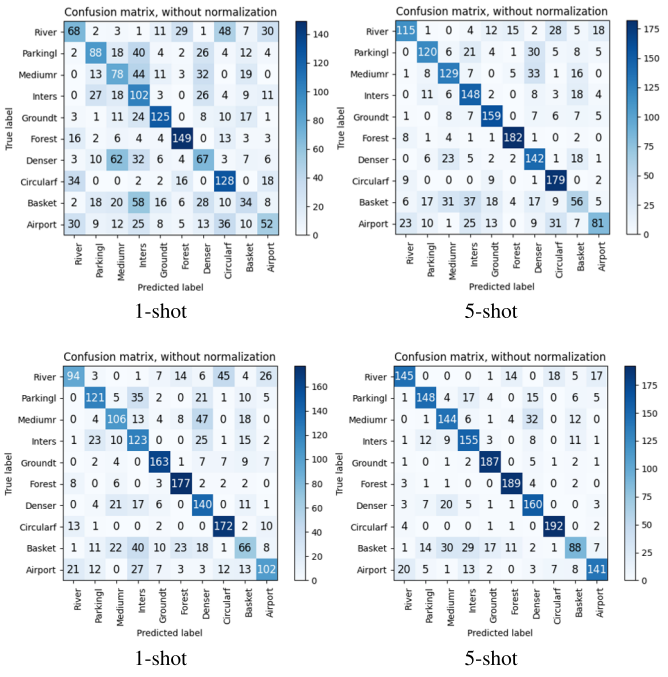

Fig. 7. Confusion matrix for novel classes under the NWPU dataset. From top to bottom are baseline [17] and our method.

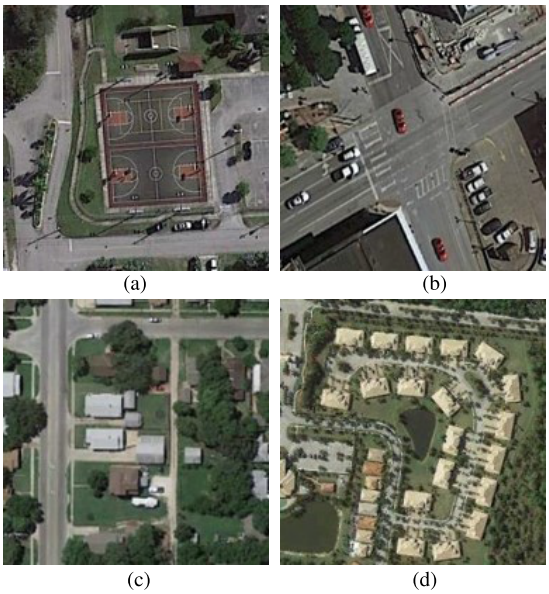

Fig. 9. Display of misclassified classes in NWPU dataset: (a) Basketball court. (b) Intersection. (c) Medium residential. (d) Dense eesidensial.

in Fig. 9, when such samples possess similar background information, classification difficulty does not decrease with an increase in sample size stemming from changes in background information.

E. Ablation Experiments

In this section, we set up ablation experiments to explore the specific role of IC-Loss, and ID-Loss in the process of sample generation. And also experimentally observe the effect of self-supervised learning for remote sensing characteristics in the feature extraction process.

The results of the ablation experiments for all classes in the three datasets under 1-shot and 5-shot tasks are shown in Table V. By comparing five settings, a 2%–6% improvement in classification accuracy is obtained by introducing SA-Loss, which indicates that designing scale-angle aware self-supervised pretext in the face of remote sensing scenes can capture better quality features. Then, after using IC-Loss or ID-Loss on top of SA-Loss, respectively. A further 1%–3% improvement is obtained in all three datasets, as it compensates for the inability of the generation model to capture diverse

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TABLE V
ABLATION EXPERIMENTS IN THREE DATASETS

SA-Loss	ID-Loss	IC-Loss	UC_Merced		NWPU		AID	
			1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
×	×	×	53.11%	68.21%	65.79%	80.85%	52.07%	66.37%
✓	×	×	58.39%$_{+5.28}$	73.08%$_{+4.87}$	67.97%$_{+2.18}$	84.26%$_{+3.41}$	56.82%$_{+4.75}$	72.87%$_{+6.50}$
✓	✓	×	59.50%$_{+6.39}$	72.97%$_{+4.76}$	70.23%$_{+4.44}$	84.95%$_{+4.10}$	59.16%$_{+7.09}$	75.58%$_{+9.21}$
✓	×	✓	59.48%$_{+6.37}$	75.47%$_{+7.26}$	68.34%$_{+2.55}$	85.10%$_{+4.25}$	57.56%$_{+5.49}$	74.07%$_{+7.70}$
✓	✓	✓	61.76%$_{+8.65}$	76.55%$_{+8.35}$	72.70%$_{+6.91}$	86.44%$_{+5.59}$	59.93%$_{+7.86}$	76.42%$_{+10.05}$

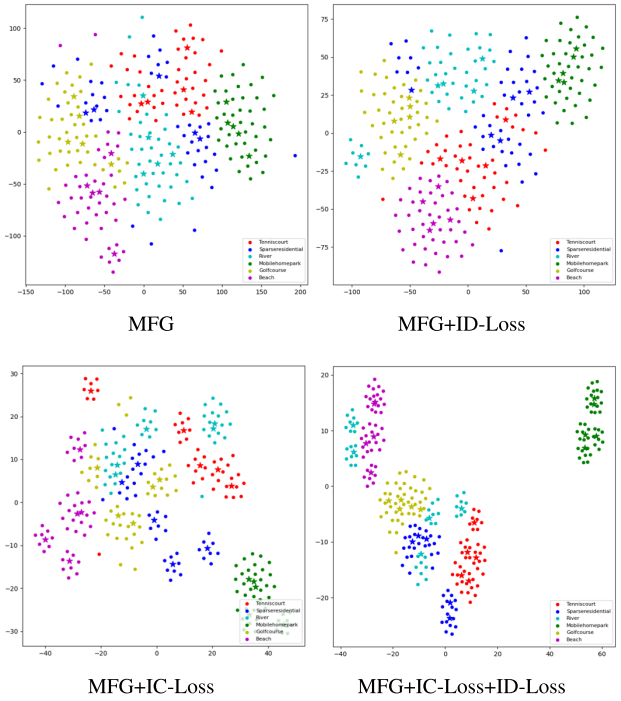

Fig. 10. Visualization of the generated sample distribution in UC_Merced. The use of IC-Loss and ID-Loss can effectively promote the discriminability and diversity of the generated samples.

intra-class features. Finally, the combined use of IC-Loss and ID-Loss reveals that the model still improves the classification accuracy by 1%–2% compared to using them alone, which indicates that these two losses have different focuses. The combined use can better facilitate the generative network to learn rich intra-class features.

We visualize the impact of IC-Loss and ID-Loss on sample generation and classification processes. As in Fig. 10, it can be intuitively felt that the effect of both losses on the diversity of the generated samples. When no loss is used in the generation process as mentioned before, the lack of the class consistency constraint leads to excessive differences between the generated samples and the original ones. Especially, the generated samples of different classes are mixed, and this situation can destroy the classification boundaries. This phenomenon is significantly alleviated by using IC-Loss, and the distance between the generated samples and the original ones is brought closer, but a certain difference is maintained between them. By adding ID-Loss on top of IC-Loss, the distribution of generated samples is more uniform and the classification boundary is clearer, moreover, the bad points are avoided. In addition, even without using IC-Loss, the generated samples

TABLE VI
ANALYSIS ON THE NUMBER OF INTRA-CLASS FEATURE GENERATION MODELS

Datasets	1	2	3	4
UC_Merced	59.50%	61.76%	**62.20%**	60.08%
NWPU	70.23%	**72.70%**	71.43%	71.11%
AID	59.16%	**59.93%**	59.75%	57.08%

are mainly concentrated around the original ones, which also proves that our method can well ensure the discriminability of the generated samples by modeling the intra-class features.

F. Parameter Analysis

1) Number of Intra-Class Feature Generation Models: In this article, MFGNet will generate multiple new samples at once to achieve data augmentation. As shown in Table VI, the impact of different numbers of intra-class feature generation models on the classification accuracy in the 1-shot task is demonstrated. If only one single branch is used, the experimental performance degrades in all three datasets due to the inability to learn the rich intra-class feature distribution. This also proves the value of the ID-Loss designed in our article. But not the more branches, the better the model performance. Through experimental observation, when using 3 or 4 branches, the model performance is not significantly improved compared to the two-branch structure. Especially in the 4-branch case, the performance in the AID dataset is not even as good as one single branch. In addition, as the number of branches increases, the number of model operations will rise. To ensure the balance between feature extraction and generation process, the number of intra-class feature generation models is set to 2.

2) Number of Sampling: In the fine-tuning stage, the number of samples generated by each intra-class feature generation model affects the performance of the classifier. An intuitive feeling is that the larger the number of samples generated, the better classification boundaries will get, but the actual situation is different from this.

The effect of different sampling numbers on the classification performance in the 1-shot task is shown in Table VII. It can be seen that the classification performance is steadily improved in the initial stage, after the number of samples reaches 4, there will be oscillations and decreases. When the generated samples can represent the intra-class features well, more sample numbers do not bring performance improvement

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TABLE VII
ANALYSIS ON THE NUMBER OF SAMPLING

Datasets	UC_Merced	NWPU	AID
0	59.97%	70.26%	57.41%
1	61.02%	71.97%	58.60%
2	61.26%	72.43%	58.05%
3	61.61%	72.14%	59.74%
4	**61.76%**	**72.70%**	**59.93%**
5	61.32%	72.90%	59.41%
6	61.48%	72.40%	59.22%
7	61.17%	73.47%	58.51%

TABLE VIII
TABLE THE EFFECT OF k ON CLASSIFICATION ACC

k	0.5	1	1.5	2	2.5
UC_Merced	59.20%	60.63%	**61.76%**	60.88%	60.85%
NWPU	70.78%	72.32%	**72.70%**	70.74%	71.95%
AID	58.23%	57.99%	**59.93%**	58.33%	58.01%

but increase the number of operations in the fine-tuning process. From the perspective of balancing the performance of models in different datasets, we require each intra-class feature generation model to be repeatedly sampled four times in the fine-tuning stage. Combined with the two-branch structure, eight new samples can be generated for each sampling.

3) Ratio of Classification Loss: In Section III, to make the network pay more attention to the generated samples during training, we introduce the parameter k. When the classification loss is calculated, the parameter k controls the proportional relationship between the classification loss of the generated samples and the original ones. As shown in Table VIII, taking the values of k as 0.5, 1, 1.5, 2, 2.5 respectively. It is obvious that either too large or too small k value will destroy the balance between the generated samples and the original ones, leading to the degradation of the classification accuracy, finally, we choose k value of 1.5.

4) Composition of Self-Supervised Learning: Based on Table V, we conducted experiments to analyze the effect of the self-supervised pretext on the model performance. The classification accuracy of the model is tested in three cases: without the self-supervised task, with the angle/scale awareness self-supervised pretext, and with the combination of two self-supervised pretexts. The results are shown in Table IX. By constructing a self-supervised pretext based on angle and scale awareness, a 3%–5% improvement in classification accuracy can be obtained. If we look at the angle and scale aware tasks separately, using a single one of them during training will have an impact on the model performance. But using the angle aware task has a greater improvement, which indicates that either the scale or the angle information will prompt the model to extract better quality features, while the angle information is more critical. In the few-shot remote sensing scene classification task, a reasonable self-supervised task to capture the diverse information of remote sensing scenes can effectively improve classification accuracy.

TABLE IX
COMPOSITION OF SELF-SUPERVISED LEARNING

Datasets	Scale-Loss	Angle-Loss	1-shot	5-shot
UC_Merced	×	×	56.85%	72.25%
	✓	×	57.70%$_{+0.85}$	73.51%$_{+1.26}$
	×	✓	59.01%$_{+2.16}$	73.75%$_{+1.50}$
	✓	✓	61.76%$_{+4.91}$	76.55%$_{+4.30}$
NWPU	×	×	68.84%	83.36%
	✓	×	69.68%$_{+0.84}$	84.17%$_{+0.81}$
	×	✓	71.05%$_{+2.21}$	85.52%$_{+2.16}$
	✓	✓	72.70%$_{+3.86}$	86.44%$_{+3.08}$
AID	×	×	56.83%	72.37%
	✓	×	57.04%$_{+0.21}$	73.47%$_{+1.10}$
	×	✓	58.48%$_{+1.65}$	74.83%$_{+2.46}$
	✓	✓	59.93%$_{+3.10}$	76.42%$_{+4.05}$

V. CONCLUSION

In this article, we propose an MFGNet to solve the few-shot problem by generating discriminative and diverse samples. MFGNet models the intra-class feature distribution and generates distinguishable samples by combining intra-class features with class-specific features. In addition, we design IC-Loss and ID-Loss at the inter-class and intra-class feature levels to ensure the variability of the generated samples and the diversity of the intra-class feature distribution, by which a clearer classification boundary can be obtained, thereby improving the classification performance. Finally, we follow the characteristics of remote sensing scenes and design an SA-Loss to help the model capture the orientation and scale information.

In the future, we will investigate the model pretraining approaches for remote sensing scenes, which can provide more discriminative intra-class features in our generation process, thereby improving the few-shot remote sensing scene classification performance.

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